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AI APTITUDE QUESTION BANK

AI for Analysts

About This Question Bank

An assessment for professionals in data and analytics roles. Tests AI-assisted data thinking, analytical workflow design, verification discipline, evidence-based communication, and judgment in applied analytical scenarios.

5	15	75
Sets	Questions per Set	Total Questions

SET GUIDE

Set 1 · Data Thinking with AI
Data thinking with AI — framing questions, analysis design, and problem setup

Set 2 · AI-Assisted Analysis
AI-assisted analysis — pattern detection, synthesis, and insight generation

Set 3 · Verification and Critical Evaluation
Verification and critical evaluation — checking AI outputs for data accuracy

Set 4 · Analytical Communication
Analytical communication — presenting AI-assisted findings professionally

Set 5 · Judgment Scenarios
Judgment scenarios — applied analyst decisions with AI involvement

Each question has one correct answer. Select the best answer from the options provided.
Correct answers and explanations follow each question.

SET
1

Data Thinking with AI

Set 1 of 5 · 15 Questions · AI for Analysts

Q1. A manager gives you a dataset and says "find insights." Before running any AI analysis, your first step is:

A. Clean and format the data so AI can process it correctly.

B. Define a specific business question the analysis should answer.

C. Run exploratory AI analysis to surface questions to investigate.

D. Identify which AI tool is most appropriate for this data type.

Correct Answer: B

Explanation

Analysis without a defined question produces random outputs. A specific business question determines what counts as an insight, what analysis is relevant, and what success looks like. Cleaning, exploring, and tool selection all come after the question is defined.

Q2. You are designing an AI-assisted analysis of customer churn. Which variable is most important to define first?

A. The AI model that performs best on classification tasks.

B. The size of the dataset needed for statistical significance.

C. The visualisation format for presenting findings to stakeholders.

D. The definition of "churn" for this specific business context.

Correct Answer: D

Explanation

"Churn" can mean different things in different businesses — cancelled subscriptions, inactivity for X days, non-renewal at contract end. Without a precise definition, the analysis answers the wrong question no matter how well the AI performs.

Q3. AI is asked to identify "trends" in your sales data. What is the most important instruction to include?

A. Specify the time period, metric, and what constitutes a meaningful trend.

B. Ask AI to identify all trends and patterns it can find in the data.

C. Specify the visualisation format for displaying the trends.

D. Ask AI to compare this dataset to industry benchmark data.

Correct Answer: A

Explanation

"Trends" is undefined without a time period, specific metric, and threshold for what is meaningful versus noise. Without these specifications, AI will surface everything — and distinguishing real trends from random variation requires your judgment, not AI's.

Q4. Which of these analyst tasks benefits most from AI assistance?

A. Making the final judgment on which hypothesis is correct.

B. Collecting primary research data from the market.

C. Generating multiple hypotheses quickly to structure the analysis.

D. Deciding which business decisions follow from the findings.

Correct Answer: C

Explanation

Hypothesis generation — rapidly exploring multiple possible explanations for a pattern — is where AI's breadth and speed adds the most value to analysis. Final judgment, data collection, and strategic decisions all require human accountability AI cannot supply.

Q5. Before asking AI to analyse your dataset, the most important data quality check is:

A. Whether the dataset is large enough to produce statistically significant results.

B. Whether the data definitions and categories are consistent throughout.

C. Whether the AI tool you are using can handle the file format.

D. Whether the dataset is recent enough to be relevant to your question.

Correct Answer: B

Explanation

Inconsistent definitions — "region" meaning different things across rows, "revenue" calculated differently for different periods — produce fundamentally misleading analyses regardless of AI capability. This is the root data quality issue before any other check.

Q6. You have two datasets on the same customer cohort from different internal systems. They disagree on customer count by 8%. Your first action is:

A. Investigate the source and definition differences between the two systems.

B. Average the two counts as a reasonable estimate.

C. Use the larger count — it is more likely to be complete.

D. Use the smaller count — it is more likely to exclude duplicates.

Correct Answer: A

Explanation

8% disagreement between datasets on the same subject is a significant data quality issue that must be diagnosed before any analysis. The investigation may reveal different customer definitions, collection periods, or filtering logic — all of which affect analytical validity.

Q7. An analyst wants to use AI to build a segmentation of customers. The most important input to define first is:

A. The number of segments that will be most manageable for the business.

B. The AI clustering algorithm best suited to customer data.

C. The data fields available for the AI to use in clustering.

D. The purpose of the segmentation — what decisions it will inform.

Correct Answer: D

Explanation

Segmentation design should start with purpose — what decisions will the segments enable? A segmentation designed for product targeting differs from one designed for service personalisation. Without this, you get technically correct clusters that serve no business need.

Q8. You feed raw survey data into Claude and ask it to find the key themes. The most important limitation to be aware of is:

A. Claude cannot process survey data with more than 100 respondents.

B. Claude will only identify themes present in its training data.

C. Claude may interpret open-text responses through its own pattern biases.

D. Claude requires structured data format to analyse survey responses.

Correct Answer: C

Explanation

When Claude identifies themes in text data, it applies its trained pattern-matching to interpret responses. This can introduce biases — prioritising certain framings, grouping responses that are actually different, or missing culturally specific meanings. Human review of theme definitions is essential.

Q9. What is the most reliable approach when using AI to size a market?

A. Define your bottom-up assumptions, use AI to stress-test them, verify inputs from primary sources.

B. Ask AI for the market size and use it as your baseline figure.

C. Use AI to aggregate publicly available market research reports.

D. Ask AI to identify the methodology used in industry reports.

Correct Answer: A

Explanation

Market sizing requires a bottom-up logic with verifiable assumptions at each step. AI can help stress-test reasoning and surface assumptions you missed — but the inputs must come from verified primary sources, not AI training data, which may be outdated or fabricated.

Q10. A stakeholder asks for "the key drivers of revenue growth." Before building the AI analysis, you should:

A. Ask AI to identify all statistically significant revenue variables.

B. Run the analysis first and define "drivers" from the findings.

C. Collect more data to ensure the analysis is comprehensive enough.

D. Clarify whether they want correlation analysis or causal attribution.

Correct Answer: D

Explanation

Correlation and causation require different analytical approaches and produce different conclusions. "Drivers" can mean correlated variables or causal factors — the difference determines methodology, data requirements, and what the stakeholder can actually act on.

Q11. Which analytical question is least appropriate for AI to answer independently?

A. Which date range in this dataset shows the highest variance.

B. Whether this data pattern represents a real business problem worth solving.

C. Which columns in this dataset have the most missing values.

D. What the average order value is across these 10,000 transactions.

Correct Answer: B

Explanation

Whether a pattern is a "real business problem worth solving" requires domain knowledge, strategic context, and judgment about organisational priorities — none of which AI has. The other three are factual, computation-based tasks that AI handles reliably.

Q12. You are about to brief senior leadership using AI-assisted analysis. What is non-negotiable before the brief?

A. The presentation design must meet the company's brand standards.

B. The AI must have been used only for approved analysis tasks.

C. Every specific figure cited must be verified from a primary source.

D. A written summary of AI's limitations must be disclosed to leadership.

Correct Answer: C

Explanation

In senior leadership briefings, every specific figure you cite is your professional assertion. AI-generated figures that turn out to be hallucinated in front of leadership create serious credibility damage. Verification is non-negotiable regardless of other considerations.

Q13. What does "framing the analysis" mean for an AI Analyst?

A. Selecting the chart types and layout for presenting findings.

B. Writing the narrative around AI-generated charts and tables.

C. Deciding which insights from the AI analysis to include in the report.

D. Defining the question, hypothesis, scope, and success criteria before starting.

Correct Answer: D

Explanation

Framing precedes analysis — it is the intellectual work of defining what question is being answered, what hypothesis is being tested, what data is in scope, and what a useful answer looks like. Without framing, AI analysis produces outputs with no clear purpose.

Q14. An AI analysis identifies a correlation between two variables. The most responsible conclusion to draw is:

A. One variable is causing the other; this is an actionable finding.

B. These variables move together; causation requires further investigation.

C. The correlation is meaningful only if it exceeds 0.7 coefficient.

D. Correlation in AI analysis confirms causation for practical purposes.

Correct Answer: B

Explanation

Correlation is not causation — and AI cannot establish causation without properly designed experiments or causal analysis methods. Presenting a correlational finding as causal leads to wrong decisions. "These move together" is the correct, responsible framing.

Q15. What is an analyst's primary responsibility when using AI for data analysis?

A. Owning the quality and validity of the conclusions, regardless of AI's role.

B. Ensuring the AI tool used is approved for data analysis tasks.

C. Documenting which parts of the analysis were AI-assisted.

D. Ensuring the AI can access all the data it needs to analyse.

Correct Answer: A

Explanation

The primary responsibility is analytical accountability — you own the conclusions whether AI did the computation or not. Approval, documentation, and data access are supporting operational requirements, not the core professional obligation.

SET
2

AI-Assisted Analysis

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Q16. You use Claude to summarise 50 customer feedback responses. Before using the summary, you should:

A. Have a second AI tool verify the summary for completeness.

B. Check whether the summary matches the original response count.

C. Read a sample of raw responses to validate the summary's accuracy.

D. Ensure Claude has not excluded responses in negative sentiment.

Correct Answer: C

Explanation

AI text summarisation can omit, distort, or misrepresent responses — especially nuanced, ambiguous, or culturally specific feedback. Spot-checking a representative sample of raw responses against the summary is the only reliable validation.

Q17. Which step in a data analysis workflow does AI most reliably accelerate?

A. Structuring and reformatting raw data into a consistent schema.

B. Making the final decision on which findings are strategically important.

C. Designing the sampling methodology for the research study.

D. Interpreting what findings mean for the organisation's specific context.

Correct Answer: A

Explanation

Structural, reformatting tasks — converting messy data into a consistent schema, standardising fields, removing formatting inconsistencies — are where AI adds the most reliable speed advantage with lowest risk. Strategic and contextual judgment tasks remain human responsibilities.

Q18. What is the most effective way to use AI to explore a dataset you have never seen before?

A. Ask AI to run a comprehensive analysis and present its top 10 findings.

B. Ask AI to clean and normalise the data before any exploration.

C. Ask AI to compare this dataset to similar industry datasets it knows.

D. Ask AI to describe the data structure, field names, and samples to orient yourself.

Correct Answer: D

Explanation

The first step with an unfamiliar dataset is orientation — understanding what fields exist, what they contain, and what the data structure looks like. This gives you the basis to form meaningful questions. Jumping to comprehensive analysis before understanding the data produces unfocused outputs.

Q19. You use AI to segment customers into 5 clusters. The most important next step before presenting the segments is:

A. Run the same clustering with 4 and 6 clusters to compare the results.

B. Interpret each cluster using your business knowledge to validate they make sense.

C. Ask AI to label each cluster with a descriptive name for presentation.

D. Calculate the statistical significance of the cluster separation.

Correct Answer: B

Explanation

AI clusters are mathematical groupings — they need human business interpretation to determine whether they represent meaningful, actionable distinctions. A cluster that is statistically distinct may be analytically meaningless if it doesn't correspond to a real business category.

Q20. You use Claude to generate the initial Python code for a data analysis pipeline. The most important next step is:

A. Run the code on the full dataset to check if it produces expected outputs.

B. Ask Claude to review and debug the code before you run any tests.

C. Review and test the code on a small data sample before running it on the full dataset.

D. Add your name to the code as author before sharing with the team.

Correct Answer: C

Explanation

AI-generated code can contain bugs, incorrect logic, or assumptions that fail on real data. Testing on a small, controllable sample first allows you to catch errors without corrupting a full analysis run.

Q21. Which AI use case in analytics carries the highest verification burden?

A. AI-generated first drafts of analyst commentary for internal review.

B. AI-assisted reformatting of raw data into a structured table.

C. AI-generated chart labels and titles for a presentation.

D. AI-generated statistics used in a board-level report.

Correct Answer: D

Explanation

Statistics cited in a board-level report are directly used for executive decisions and are typically shared externally. Errors here have the highest professional and organisational consequences. Internal drafts, formatting, and labels can be corrected before they matter.

Q22. An AI tool tells you that a metric "increased significantly." What is your first analytical response?

A. Ask: significant compared to what baseline, over which period, by what magnitude?

B. Accept the finding and include it in your report as a positive trend.

C. Ask the AI to confirm this finding with a second statistical test.

D. Investigate whether the underlying data collection method changed.

Correct Answer: A

Explanation

"Significant" in AI-generated analysis is often used loosely — it may mean large in absolute terms, statistically significant, or just noteworthy by AI's judgment. Unpacking the comparison, timeframe, and magnitude turns a vague observation into an analytical fact.

Q23. You run the same analysis prompt twice on the same dataset and get slightly different numerical outputs. What does this indicate?

A. The dataset has inconsistent data that produces variable results.

B. Temperature variation is producing non-deterministic outputs — use temperature = 0.

C. The AI is selecting different analysis methods in each session.

D. One run had a context window overflow that truncated the analysis.

Correct Answer: B

Explanation

Numerical analysis tasks require deterministic outputs — the same input should always produce the same result. Using temperature = 0 eliminates sampling randomness and is essential for any analysis task where consistency and reproducibility matter.

Q24. A stakeholder says "the AI found this pattern — it must be true." The correct challenge is:

A. "AI findings are accurate by default for pattern detection tasks."

B. "This finding is preliminary — we should run it through a second AI tool."

C. "AI pattern detection is reliable for well-structured datasets."

D. "AI identified this correlation — we still need to validate it with additional analysis."

Correct Answer: D

Explanation

AI identifies patterns statistically — which means it can find spurious correlations, surface noise, or flag patterns that are not robust. Every AI-identified pattern is a hypothesis to be validated with further analytical work, not a confirmed finding.

Q25. What is the most important difference between AI "analysing" data and a human analyst doing so?

A. The human is slower but more accurate in numerical computation.

B. The AI processes larger datasets than a human could review manually.

C. The human applies contextual business judgment to interpret what patterns mean.

D. The human can access primary sources that AI cannot reach.

Correct Answer: C

Explanation

AI processes patterns statistically; a human analyst interprets what those patterns mean for this specific business in this specific context with this specific strategy. That interpretive, contextual layer is what converts data outputs into business insight.

Q26. You need to present AI-assisted analysis to a client who is unfamiliar with AI. The most important thing to communicate is:

A. Which AI tool was used and its technical accuracy rating.

B. Which parts of the analysis were AI-generated and how they were validated.

C. That AI was not used for any subjective or interpretive conclusions.

D. That the findings would be the same whether AI was used or not.

Correct Answer: B

Explanation

Clients deserve to understand the production method for the analysis they are acting on — specifically where AI was involved and how those outputs were validated. This enables informed judgment about the confidence level of specific findings.

Q27. You want to use AI to build an analyst's daily reporting workflow. The first workflow step must be:

A. Data validation — confirming the incoming data is complete and consistent.

B. AI analysis — generating the daily metrics and trend commentary.

C. Visualisation — converting AI outputs into charts for the report.

D. Distribution — sending the report to the relevant stakeholders.

Correct Answer: A

Explanation

Data validation is always the first step in any reporting workflow — garbage in, garbage out. If the incoming data is incomplete, duplicated, or inconsistently formatted, every downstream AI output is wrong. No amount of AI sophistication recovers from bad input data.

Q28. What is the most important skill an analyst develops by working with AI on data tasks?

A. The ability to write Python and SQL to extract data for AI processing.

B. The speed to process and present more analysis than without AI.

C. The judgment to direct AI toward the right question and evaluate its outputs.

D. The ability to use the widest range of AI analytics tools available.

Correct Answer: C

Explanation

AI amplifies the analyst's core judgment skill — the ability to ask the right question, direct the analysis appropriately, and critically evaluate outputs against business reality. Speed, technical skills, and tool breadth are secondary to this fundamental judgment capability.

Q29. An AI analysis of your customer data produces a finding you strongly disagree with. What is the right response?

A. Accept the AI finding — your intuition may be biased.

B. Reject it and redo the analysis until it matches your expectation.

C. Present both the AI finding and your intuition and let the stakeholder decide.

D. Investigate the data and methodology before either accepting or rejecting it.

Correct Answer: D

Explanation

Disagreement between AI output and domain expertise is diagnostic, not decisive either way. Investigate: is the data correct? Is the analysis method right? Is your prior belief based on outdated information? The investigation determines who was right.

Q30. What does "analytical integrity" mean when AI is involved in the analysis?

A. Using only AI-verified data and AI-validated methods.

B. Being transparent about AI's role and accountable for the conclusions.

C. Ensuring AI was used only for approved analytical task types.

D. Documenting every AI interaction in the analysis audit trail.

Correct Answer: B

Explanation

Analytical integrity means being transparent about how the analysis was done (including AI's role) and being fully accountable for the conclusions (regardless of how they were produced). Transparency and accountability are the core principles; documentation and approval are supporting practices.

SET
3

Verification and Critical Evaluation

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Q31. An AI tool presents a chart showing revenue increasing 40% year-on-year. Before presenting it, you should:

A. Verify the underlying data and confirm the calculation methodology.

B. Check whether the chart design meets your presentation standards.

C. Ask the AI to confirm the finding is statistically significant.

D. Compare the 40% figure to the company's own growth targets.

Correct Answer: A

Explanation

Any specific figure — 40% — must be traceable to the underlying data with a clear calculation method. AI can generate incorrect figures that look correct in a chart. Verification precedes presentation regardless of how compelling the visual looks.

Q32. You use AI to draft a competitive analysis. The section you must verify most carefully is:

A. The analytical framework used to structure the comparison.

B. The section headings and narrative flow of the document.

C. Specific claims about competitors' products, pricing, or performance.

D. The summary recommendations at the end of the analysis.

Correct Answer: C

Explanation

Competitor-specific facts — product capabilities, pricing, market share, strategic moves — are exactly where AI hallucination risk is highest for competitive analysis. AI may confidently state wrong or outdated competitor information. Verify every specific competitor claim.

Q33. What is the most effective prompt technique for reducing hallucination in AI-generated data analysis?

A. Ask AI to cite its confidence level for each analytical finding.

B. Instruct AI to work only from the provided dataset and flag when it cannot.

C. Run the analysis 3 times and compare outputs for consistency.

D. Use the most advanced model available for all analytical tasks.

Correct Answer: B

Explanation

Grounding instructions — explicitly directing Claude to use only the provided data and to flag when it is going beyond that data — is the most direct way to reduce hallucination in analysis. Confidence levels are unreliable; repeated runs just produce more potentially hallucinated outputs.

Q34. An AI analysis identifies an anomaly in your data. What is the first question to ask?

A. "Is this anomaly large enough to report to stakeholders?"

B. "Which AI model detected this anomaly and how reliable is it?"

C. "What is the recommended action for this type of anomaly?"

D. "Is this a data quality issue or a real business phenomenon?"

Correct Answer: D

Explanation

Anomalies in data are often data quality artifacts — duplicate entries, collection errors, system migrations — rather than real business events. Before investigating the business implication, determine whether the anomaly reflects reality or a data problem.

Q35. You receive an AI summary of 100 customer reviews that says "customers are generally satisfied." You want to probe this further. The most effective next step is:

A. Read a random sample of negative reviews to test the summary's accuracy.

B. Ask AI to re-summarise with a focus on negative sentiment only.

C. Filter the raw reviews to those with the lowest star ratings.

D. Ask AI to list specific complaints mentioned more than five times.

Correct Answer: A

Explanation

"Generally satisfied" can mask important variation. Reading a random sample of negative reviews tests whether the AI summary is accurate or whether it is averaging away important dissatisfaction signals that matter for product decisions.

Q36. Which of these is the strongest evidence that an AI-generated analysis is reliable?

A. The analysis was generated by the highest-rated AI analytics tool.

B. Every specific claim traces to a verifiable primary source.

C. The output was consistent across 5 separate generation runs.

D. The analysis structure matches what a senior analyst would produce.

Correct Answer: B

Explanation

Traceability to primary sources is the only evidence that actually demonstrates reliability. Tool ratings are marketing; consistency across runs means the same hallucination repeated; structural quality is no indicator of factual accuracy.

Q37. A colleague presents AI-generated market research numbers without citing sources. Your response is:

A. "These numbers look reasonable, so let's include them in the analysis."

B. "AI market research is generally accurate enough for strategic planning."

C. "We should re-run this with a different AI tool to cross-check."

D. "We need to verify these before using them — can you show me the source data?"

Correct Answer: D

Explanation

"Looks reasonable" is not verification — it is pattern-matching to your prior beliefs. Every figure used in strategic analysis needs a traceable source. The professional response is to request the source data before the numbers are used.

Q38. You use AI to run sentiment analysis on employee survey data. The finding shows 72% positive sentiment. What is the most important verification step?

A. Compare the 72% figure to industry benchmarks for employee sentiment.

B. Run the same data through a second sentiment analysis tool.

C. Review how AI defined "positive" and check a sample of categorised responses.

D. Ask HR whether 72% aligns with their own qualitative observations.

Correct Answer: C

Explanation

Sentiment analysis labels depend on how "positive" is defined by the AI model — and that definition may not align with your organisation's context. Reviewing the definition and spot-checking categorised responses validates whether the 72% is measuring what you think it is.

Q39. What is the most common error analysts make when evaluating AI-generated statistical claims?

A. Not verifying whether the calculation method was disclosed.

B. Accepting the number without checking what population it describes.

C. Not running a second statistical test to confirm the result.

D. Presenting the number without rounding it appropriately.

Correct Answer: B

Explanation

AI-generated statistics often lack a clearly defined population — "72% of customers" may mean 72% of paying customers, active users, trial users, or some undefined sample. The population definition changes everything about what the number means and whether it is usable.

Q40. Before using an AI-generated cohort analysis in a strategy document, what must you confirm?

A. The cohort definition and whether it is consistent across all time periods.

B. Whether the AI tool is approved for strategic analysis tasks.

C. Whether the cohort size is large enough for statistical significance.

D. Whether the analysis was generated by the latest model version.

Correct Answer: A

Explanation

Cohort analysis validity depends entirely on consistent cohort definition across time. If "June 2024 cohort" means different things at different time points in the analysis, the entire comparison is invalid regardless of sample size or model version.

Q41. You are producing a quarterly business review and some figures were AI-generated. What is the minimum standard before the document is distributed?

A. The document notes which figures were AI-generated for disclosure.

B. An AI proofreading tool has checked the document for consistency.

C. Every AI-generated figure is verified against the source data by a human.

D. The AI-generated figures are flagged with a lower confidence interval.

Correct Answer: C

Explanation

The minimum standard is verification — a human checks every AI-generated figure against the actual source data before the document is distributed. Disclosure, proofreading tools, and confidence flags do not substitute for verification.

Q42. A stakeholder challenges a finding from your AI-assisted analysis. What is the strongest response?

A. "I used the most advanced AI model available for this analysis."

B. "The AI is highly accurate on this type of data task."

C. "I can re-run the analysis to confirm it produces the same result."

D. Walk through the source data and methodology step by step.

Correct Answer: D

Explanation

The only credible defence of an analytical finding is methodological transparency — showing the source data and the steps used to reach the conclusion. Model prestige, AI accuracy claims, and replication do not address the stakeholder's substantive challenge.

Q43. You discover an error in data you already used for an AI analysis in last quarter's report. What do you do?

A. Assess whether the error would change the conclusion before disclosing.

B. Correct the data, re-run the analysis, and disclose the correction to stakeholders.

C. Correct it for future reports but do not revise the distributed document.

D. Check whether anyone has used the previous report's figures before acting.

Correct Answer: B

Explanation

Analytical integrity requires disclosing corrections regardless of whether the conclusion changes. Stakeholders may have made decisions based on the incorrect figures. The correction, disclosure, and corrected analysis are all required regardless of materiality.

Q44. What distinguishes a high-quality AI-assisted analyst from a low-quality one?

A. The high-quality analyst uses more AI tools in their analysis workflow.

B. The high-quality analyst relies on AI for a higher proportion of their work.

C. The high-quality analyst treats AI outputs as hypotheses requiring validation.

D. The high-quality analyst discloses AI use in all analytical outputs.

Correct Answer: C

Explanation

The defining characteristic is the hypothesis mindset — treating every AI output as a starting point that requires validation rather than a finding to be presented. This is the intellectual standard that separates reliable AI-assisted analysis from dangerous AI-dependent analysis.

Q45. An AI flags a finding as "statistically significant." What analytical caution should you apply?

A. Statistical significance depends on sample size and does not guarantee practical importance.

B. Statistical significance from AI is more reliable than from manual testing.

C. Statistical significance means the finding is correct and ready to present.

D. Statistical significance requires confirmation from a domain expert.

Correct Answer: A

Explanation

Statistical significance is a mathematical property that depends heavily on sample size — with large enough samples, trivially small effects become statistically significant. Statistical significance does not mean practically important, clinically meaningful, or strategically relevant.

SET
4

Analytical Communication

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Q46. You need to present a complex AI-assisted analysis to a non-technical executive. The most important principle is:

A. Explain the AI methodology so the executive understands the basis.

B. Show the data first so the executive can draw their own conclusions.

C. Include confidence intervals so the executive understands uncertainty.

D. Lead with the business implication, not the analytical method.

Correct Answer: D

Explanation

Executives need to make decisions — the primary communication should be what the finding means for the business and what action it suggests. Methodology, raw data, and uncertainty quantification are supporting context, not the lead.

Q47. A colleague drafts an AI-assisted report that presents every finding at the same level of confidence. The problem is:

A. Presenting all findings at equal confidence makes the report too long.

B. Different findings have different evidence quality and should be differentiated.

C. Executive readers will not know which findings are from AI.

D. A uniform confidence level violates analytical communication standards.

Correct Answer: B

Explanation

Different findings in a report have different levels of evidence — verified from primary sources, inferred from patterns, or extrapolated from proxies. Presenting them at the same confidence level misleads readers about which conclusions are solid and which are speculative.

Q48. You use Claude to write an executive summary of your analysis. The summary sounds compelling but omits two findings that contradict the main narrative. What must you do?

A. Ask Claude to rewrite the summary with a more balanced narrative.

B. Present the summary as written and include contradicting findings in an appendix.

C. Revise the summary to include the contradicting findings honestly.

D. Note in the summary that AI generated it, so the reader can judge independently.

Correct Answer: C

Explanation

Omitting contradicting findings is analytical dishonesty — it distorts the evidence and misleads decisions. The summary must represent the evidence accurately, including findings that complicate the narrative. An appendix placement is still selective presentation.

Q49. What is the most effective structure for communicating an AI-assisted data finding?

A. Finding — Evidence — Implication — Confidence level

B. Method — Data — Finding — Recommendation

C. Context — Analysis — Charts — Conclusion

D. Background — AI tool used — Output — Next steps

Correct Answer: A

Explanation

The most effective analytical communication structure leads with the finding (what was discovered), supports it with evidence (what data shows it), explains the implication (why it matters), and is honest about confidence (how certain we are). This matches how executives consume analysis.

Q50. An AI-generated visualisation shows a trend that looks dramatic but the Y-axis starts at 94% rather than 0%. What should you do?

A. Keep the chart as-is — it correctly shows the relevant range of variation.

B. Add a footnote explaining that the Y-axis does not start at zero.

C. Ask AI to regenerate the chart with a different visualisation type.

D. Reconstruct the chart with a Y-axis starting at 0 to show true scale.

Correct Answer: D

Explanation

A Y-axis that does not start at zero exaggerates the apparent magnitude of change. For trend data that should be presented in context of its true scale, the Y-axis should start at 0. A footnote acknowledges but does not fix the visual distortion.

Q51. You present AI-generated analysis and a stakeholder asks a question you cannot answer. The correct response is:

A. "The AI produced this analysis — you would need to ask it directly."

B. "The AI is confident in this finding, so the question may not apply."

C. "I will investigate and get back to you with a verified answer."

D. "Let me re-prompt Claude right now to see if it can answer."

Correct Answer: C

Explanation

You own the analysis — not the AI. "I'll investigate and verify" is the professional response. Directing the stakeholder to the AI, appealing to AI's confidence, or live-prompting in a presentation all undermine your credibility and the quality of the answer.

Q52. What is the correct way to handle uncertainty in an AI-assisted analytical report?

A. State clearly which conclusions are high-confidence and which are directional.

B. Remove all findings with uncertainty from the final report.

C. Present all findings with confidence intervals calculated by the AI.

D. Disclose that AI was used and ask the reader to judge uncertainty.

Correct Answer: A

Explanation

Analytical reports serve decision-makers best when they clearly distinguish high-confidence conclusions (based on verified data) from directional hypotheses (based on patterns requiring further validation). Removing uncertain findings or outsourcing uncertainty judgment to the reader does not serve the reader.

Q53. Which element of an AI-assisted analysis is most important to attribute clearly in a report?

A. The name and version of the AI tool used for the analysis.

B. Which specific claims were generated by AI versus verified from primary data.

C. The total number of AI interactions in the analysis process.

D. Which team member was responsible for the AI prompting work.

Correct Answer: B

Explanation

The most important attribution is at the claim level — which specific facts and figures came from AI generation and which were verified from primary sources. This tells the reader exactly how to calibrate confidence in each part of the analysis.

Q54. A stakeholder says your AI-assisted analysis is "just opinion." The strongest professional response is:

A. "This analysis used the most advanced AI tool currently available."

B. "The AI model processes far more data than human analysis could."

C. "Here are the verified primary sources each key finding is based on."

D. "I have extensive experience in this domain so the interpretation is sound."

Correct Answer: C

Explanation

The only effective response to an accusation of opinion is evidence — specifically, the primary sources that each finding is based on. Tool prestige, data volume, and personal expertise are not evidence; traceable sources are.

Q55. You want to use AI to write the narrative sections of your analysis report. The most important instruction to include in the prompt is:

A. "Write in a formal, professional tone suitable for executive readers."

B. "Ensure the narrative highlights the most positive findings first."

C. "Generate narrative of approximately 300 words per section."

D. "Do not state or imply anything that is not explicitly supported by the findings below."

Correct Answer: D

Explanation

The most important instruction is the grounding constraint — directing Claude to write only what the actual findings support. Without this, AI will extrapolate, add context from its training data, and generate narrative claims that go beyond what the analysis actually shows.

Q56. An AI-generated analysis uses precise-sounding figures from its training data rather than from your dataset. The most likely cause is:

A. The prompt did not sufficiently specify that AI should use only the provided data.

B. The AI model cannot distinguish between its training data and provided datasets.

C. The dataset was too small for the AI to use as a primary source.

D. The AI tool being used requires explicit API configuration to limit data sources.

Correct Answer: A

Explanation

Without clear grounding instructions, AI will blend your dataset with its own training knowledge — filling gaps with training data statistics that sound plausible but are not from your analysis. Explicit data grounding is required in every analytical prompt.

Q57. What is the most professional way to communicate that AI was used in your analysis?

A. Add a general disclaimer noting "AI tools were used in this analysis."

B. State specifically which tasks AI assisted with and how outputs were verified.

C. Only disclose if the client or audience directly asks about your process.

D. AI disclosure is unnecessary when outputs have been fully verified.

Correct Answer: B

Explanation

Meaningful disclosure is specific — stating what AI did (generated first draft hypotheses, structured the dataset, drafted narrative sections) and how those outputs were validated. A generic disclaimer is too vague to be informative; non-disclosure assumes AI use is invisible.

Q58. You are presenting analysis to a board that is unfamiliar with AI. Which statement best introduces your methodology?

A. "AI replaced the manual analysis process, producing faster and more accurate outputs."

B. "We ran this analysis using the latest AI model to ensure accuracy."

C. "AI was used throughout, but human judgment was applied at every decision point."

D. "We used AI to structure and accelerate the analysis, with all key findings verified from primary data."

Correct Answer: D

Explanation

"Structure and accelerate, with key findings verified" is the most accurate and reassuring framing — it acknowledges AI's role, clarifies the value (speed, structure), and is honest that verification was required. The other options either overclaim accuracy or are too vague about what verification occurred.

Q59. A junior analyst on your team presents AI outputs directly as analysis findings without review. What is the correct response?

A. Explain why AI outputs are starting points and establish a review process.

B. Accept the findings since the analyst will learn from any errors over time.

C. Re-run the analysis yourself to check whether the outputs are accurate.

D. Flag the concern to your manager before addressing it with the analyst.

Correct Answer: A

Explanation

The right response is immediate, direct, and developmental — explain why AI outputs require review, and establish a clear quality process. Allowing errors to accumulate "as learning," re-running it yourself without addressing the behaviour, or escalating before addressing it directly are all suboptimal.

Q60. What makes analytical communication AI-native rather than AI-assisted?

A. The analysis was produced entirely by AI without human editing.

B. AI tools are used for both analysis and the final presentation design.

C. AI is used to accelerate production of the full analysis and verification is built into the workflow.

D. The analysis workflow was designed by an AI tool for efficiency.

Correct Answer: C

Explanation

AI-native analytical communication means AI is embedded in how the analysis is produced from end to end — not just used for one step. The key is that verification and quality control are built into the workflow, not bolted on as an afterthought.

SET
5

Judgment Scenarios

Set 5 of 5 · 15 Questions · AI for Analysts

Q61. You are under time pressure and must use AI to complete an analysis quickly. The most important principle to maintain is:

A. Verify the key figures even if you must abbreviate the written commentary.

B. Complete the full analysis and skip verification to meet the deadline.

C. Use AI for the full analysis and disclose it was produced under time pressure.

D. Deliver partial verified analysis rather than complete unverified analysis.

Correct Answer: A

Explanation

Time pressure does not eliminate verification responsibility — it requires triage. Verifying the key figures (the ones decisions will be made on) and abbreviating other elements is the right trade-off. Delivering partial but verified analysis is better than complete but unreliable analysis.

Q62. Your AI analysis contradicts a widely held view in your organisation. What is the correct response?

A. Soften the finding to reduce organisational friction.

B. Align the analysis with the prevailing view to ensure adoption.

C. Bury the finding in an appendix to avoid unnecessary conflict.

D. Validate the analysis rigorously before presenting it — then present it clearly.

Correct Answer: D

Explanation

Analytical integrity requires presenting valid findings even when they are uncomfortable. But the standard of rigour applied before presenting a challenging finding should be higher — ensure the analysis is bulletproof before you bring a finding that challenges consensus.

Q63. A client asks you to use AI to predict their churn rate for next quarter. What is the most honest response?

A. "AI can accurately predict churn using your historical data."

B. "AI can model patterns from past behaviour, but predicting the future requires assumptions we should define together."

C. "AI prediction is not reliable enough for forward-looking business metrics."

D. "I can produce an AI prediction, but it should be treated as directional only."

Correct Answer: B

Explanation

The honest response defines both what AI can do (model historical patterns) and what it cannot do (guarantee future prediction). Inviting the client to co-define the assumptions establishes that the prediction is a model with explicit inputs, not a black-box forecast.

Q64. You discover mid-project that the data you have been using for AI analysis was mislabelled. What do you do?

A. Continue the analysis and note the mislabelling as a limitation.

B. Re-run only the analyses that will be presented to the client.

C. Stop the analysis, correct the data labels, and re-run all affected analyses.

D. Assess the magnitude of the impact before deciding whether to re-run.

Correct Answer: C

Explanation

Mislabelled data invalidates all downstream analyses — there is no partial fix. The correct response is to stop, correct, and re-run. Continuing or selectively re-running produces analysis built on a known flaw, which is not analytically defensible.

Q65. You use AI to build a model that predicts customer lifetime value. The model shows 85% accuracy on test data. Before deploying it, the most important check is:

A. Whether the test data represents the full diversity of real production scenarios.

B. Whether 85% accuracy is above the industry benchmark for this model type.

C. Whether the model was tested on data the AI had not seen during training.

D. Whether the accuracy has been confirmed by a second AI model.

Correct Answer: A

Explanation

Test set representativeness is the critical question — if the test data is not representative of the full range of real-world cases (different customer segments, edge cases, time periods), 85% accuracy on the test set does not predict 85% accuracy in production.

Q66. A senior stakeholder asks you to "just get AI to do the analysis" without specifying what question to answer. What do you do?

A. Run exploratory AI analysis and present whatever findings emerge.

B. Use AI to identify what questions are worth answering first.

C. Clarify the question before proceeding — purposeless analysis wastes time.

D. Build a comprehensive dashboard covering all available metrics.

Correct Answer: C

Explanation

"Just use AI" without a question is not an instruction — it is a starting condition. Analysis requires a question. Clarifying the question with the stakeholder is the professional first step. Exploring without a question produces outputs the stakeholder cannot use.

Q67. An AI analysis produces three findings that clearly support a strategic decision and one finding that complicates it. What do you do?

A. Present only the supporting findings since they are more numerous.

B. Aggregate all findings into a single summary recommendation.

C. Present the complicating finding as a "risk" in a separate section.

D. Present all four findings and let the decision-maker weigh the full evidence.

Correct Answer: D

Explanation

Analytical integrity requires presenting the full picture. Selective presentation distorts the decision-maker's view of the evidence. The complicating finding may be the most important one. Present all findings and explain what they collectively suggest.

Q68. You use Claude to assist with a sensitive analysis involving confidential company data. Before doing so, you must confirm:

A. That Claude is the correct AI tool for confidential data analysis.

B. That your organisation's data policies permit use of this data with external AI tools.

C. That the data is anonymised before being shared with Claude.

D. That Claude's outputs will be kept internal and not published.

Correct Answer: B

Explanation

Sharing confidential company data with external AI services may violate data protection policies, client agreements, or regulatory requirements. Confirming policy compliance is the prerequisite — not just anonymisation or output control, which do not address the underlying permission question.

Q69. What is the right mental model for an analyst who uses AI every day?

A. AI is an expert colleague whose analysis I can rely on directly.

B. AI is a junior analyst who learns and improves with each task.

C. AI is a high-speed research and production assistant — I direct and verify.

D. AI is a database that retrieves the correct analysis for any question.

Correct Answer: C

Explanation

"High-speed assistant I direct and verify" is the most accurate mental model — it acknowledges AI's genuine speed advantage, preserves the analyst's directive and verification role, and avoids over-reliance. AI does not learn from session to session or retrieve correct answers like a database.

Q70. Your team produces an AI-assisted forecast that turns out to be significantly wrong. What is the most productive response?

A. Analyse what assumptions failed and improve the model and validation process.

B. Stop using AI for forecasting — the technology is not ready.

C. Disclose the failure to stakeholders and offer a refund for the work.

D. Accept that forecasting is inherently uncertain and move on.

Correct Answer: A

Explanation

Forecast failure is a learning opportunity — identifying which assumptions were wrong and how the model failed drives improvement. Abandoning AI for forecasting or accepting failure without analysis misses the diagnostic value that makes future work better.

Q71. A client says: "I trust your AI-assisted analysis more than your previous manual analysis because AI is objective." The correct response is:

A. "That's a reasonable view — AI removes human bias from the process."

B. "AI analysis has different failure modes than manual analysis — objectivity depends on data quality and question framing."

C. "AI-assisted analysis is indeed more reliable for pattern detection tasks."

D. "I appreciate the confidence — AI does improve consistency in our process."

Correct Answer: B

Explanation

AI is not objective — it reflects biases in its training data and the framing choices in the prompt. The honest response is to clarify that AI introduces different failure modes, not fewer. Accepting the client's characterisation sets up unrealistic expectations.

Q72. You are asked to scale up an AI-assisted analysis workflow from 10 reports per month to 200. The most important thing to address first is:

A. Which AI model tier can handle 200 reports within the API rate limits.

B. How much the AI tool licensing costs at the higher volume.

C. Whether the prompt templates can be reused across all 200 reports.

D. How quality review will scale alongside the increase in volume.

Correct Answer: D

Explanation

Volume scaling in AI workflows must be matched by quality review scaling. Without this, you produce 200 reports with the same error rate as 10 — but with 20x the volume of errors reaching clients. The review process is the constraint that must scale first.

Q73. What is the most important analyst skill that AI cannot replicate?

A. Reading large datasets quickly to identify patterns and anomalies.

B. Producing consistent analysis across large volumes of reports.

C. Understanding the organisational context that makes a finding matter.

D. Applying statistical methods accurately to structured data.

Correct Answer: C

Explanation

Organisational context — knowing the company's history, political dynamics, strategic priorities, and what findings will actually be acted on — is the knowledge that makes analysis strategically relevant. AI has no access to this internal context and cannot replicate the judgment that comes from it.

Q74. An analyst says "I used AI to check my work and it confirmed everything." What is the problem?

A. AI confirmation tools are only reliable for grammar and formatting.

B. Using AI to confirm AI-assisted work does not provide independent verification.

C. The analyst should have used a different AI tool for checking.

D. Confirmation of completed work is not a useful analytical practice.

Correct Answer: B

Explanation

Using AI to confirm AI-assisted work is circular — the same learned patterns that generated the analysis are being used to confirm it. Independent verification requires going outside the AI system to primary sources or domain expert review.

Q75. What distinguishes a Menler AI Analyst from someone who just uses AI tools for data work?

A. The Menler Analyst uses more advanced AI analytics tools than the average analyst.

B. The Menler Analyst has completed a certified AI analytics training programme.

C. The Menler Analyst works faster on analysis tasks using AI than without.

D. The Menler Analyst designs reliable analytical workflows and owns the quality of conclusions.

Correct Answer: D

Explanation

The Menler AI Analyst distinction is ownership — of workflow design, of verification standards, and of analytical conclusions regardless of how they were produced. Tool sophistication, certification, and speed are secondary to this responsibility and judgment standard.

END OF QUESTION BANK

Your turning point starts here.